DANEBEN! SMART IOT WASTE BIN SYSTEM — COMPLETE BILL OF MATERIALS (BOM)

Document ID: BOM-IOT-2026-FULL | Status: PRELIMINARY (BOM NOT FINAL) | Scope: All System Components, Carrier Board v2.0 & Enclosure | Date: October 2026

Pos.	Component & Technical Specification	Website / Vendor	Search Term / SKU / MPN	Qty	Est. Unit Price	Critical Remarks & Verification
01	Seeed Studio XIAO ESP32-S3 Sense Dual-Core 240MHz, 8MB PSRAM, 8MB Flash, OV2640 Cam, Mic,	Farnell (de.farnell.com) (Alt: Reichelt)	113991115 (Reichelt: XIAO ESP32S3 SEN)	**1×**	€14.35 (Alt: €15.00)	CRITICAL: Must be "Sense" model (MPN: 113991115). 8MB PSRAM & 8MB Flash required for Edge Impulse model.
02	Waterproof Ultrasonic Sensor Module JSN-SR04T / AJ-SR04M (IP67 sealed, 20–450cm, 40kHz, 2.5m	Funduino.com (Alt: Roboter-Bausatz / Amazon)	AJ-SR04M / JSN-SR04T (RBS: RBS11632)	**2×**	€6.21 (€12.42 for 2)	Do not cut or shorten the 2.5m coaxial cable. Plugs into carrier board docking bays.
03	MAX98357A I2S 3.2W Class-D Audio Amp Pure digital I2S, 3.2W @ 4Ω, 92% efficiency	Funduino.com (Alt: BerryBase.de)	MAX98357 Audioverstärker (BerryBase: ADA5770 / ADA3006)	**1×**	€4.08 (Alt: €4.90)	Strictly require I2S digital audio (eliminates Wi-Fi/BLE RF buzzing hum).
04	3W 4Ω / 8Ω Mini Speaker (~40 mm) Ø 40mm, 3W, 4Ω or 8Ω, -20°C to +70°C	Funduino.com (Alt: BerryBase.de) (3rd choice: Reichelt.de)	Lautsprecher 3W 4 Ohm (BerryBase: Mono-Lautsprecher 3W) (Reichelt: Ekulit AL-23P)	**1×**	€2.40 (Alt: €2.50 / Reichelt: €2.70)	Terminates directly into carrier board screw terminal J_SPK.
05	CN3791 MPPT Solar Charge Controller Board (6V) Maximum Power Point Tracking for 1S Li-Ion (3.7V/4.2V), up to	Funduino.com (Alt: BerryBase / Amazon)	CN3791 6V Solar Laderegler (Amazon: CN3791 6V)	**1×**	€4.20 (Alt: €6.90)	Must be 6V version to match solar panel Voc/Vmp.
06	5V / 6V 1.5W – 2.0W Monocrystalline Solar Panel ETFE laminated, UV-resistant, ~110×69mm to 140×88mm,	Roboter-Bausatz.de (Alt: Funduino / Amazon)	Solar Panel 5V 2W (88x142mm) (Funduino: Solarzelle 5V 1.25W)	**1×**	€9.59 (Alt: €4.90 Funduino)	Flush-mounts to 20° sloped enclosure roof.
07	Samsung INR18650-35E Li-Ion Cell (3.7V, 3400–3500 mAh) Grade-A authentic 18650 cell, low self-discharge, -20°C to +60°C	NKon.nl (Alt: BerryBase / Akkuteile)	Samsung INR18650-35E (BerryBase: NCR18650B)	**1×**	€1.95 (Alt: €7.90 BerryBase)	NKon.nl is primary wholesale source for authentic Samsung/LG cells.
08	18650 Single-Cell Battery Holder with Wire Leads 77 × 21 mm, spring contact, polarized color-coded wire leads	BerryBase.de (Alt: Reichelt.de / Amazon)	Batteriehalter 1x 18650 (Reichelt: Keystone 1042)	**1×**	€1.30 (Alt: €2.10 Reichelt)	Spring-contact clip with wire leads for battery tube.
09	IP67 Waterproof Enclosure & Gasket (~120 × 80 × 55 mm) UV-stabilized Polycarbonate/ABS, silicone sealing O-ring	Amazon.de Prime (Alt: Reichelt / Spelsberg)	IP67 Abzweigdose 115x90x55 (Reichelt: Spelsberg TK PS)	**1×**	€5.49 (Alt: €6.50 Reichelt)	Ensures complete outdoor IP67 dust and water protection.
10	Stainless Steel Hose Clamps (V2A, Ø 70–120 mm) Heavy-duty marine stainless steel screw clamps for pole	Amazon.de Prime (Alt: Baumarkt / Reichelt)	Schlauchschellen Edelstahl 70-120mm 2er	**2×**	€2.50 (for 2 clamps)	Rust-proof V2A stainless steel for municipal bin mounting.
11	Optical Glass Disc (Ø 20mm) + ePTFE Acoustic Vent Borosilicate lens window + hydrophobic breathable vent sticker	Amazon.de Prime	Optisches Glas 20mm + ePTFE Membran	**1 Set**	€4.99 (Lens & Vent Set)	Protective lens window and hydrophobic acoustic vent sticker.
12	Daneben Modular Prototype Carrier Board PCB v2.0 130×90mm FR4 2-Layer, dual sensor docks, USB-C, level-shifters	JLCPCB / PCBWay (Alt: In-House Fabrication)	DANEBEN-CARRIER-V2 (Gerbers from hardware/)	**1×**	€3.00 (Batch of 5: €15.00)	Custom fabricated motherboard housing all modules, daughterboards, and user interface.
13	Carrier Board Interface Components & Calibration Controls Dual 12×12mm buttons, slide switch, USB-C, LEDs, terminals,	BerryBase.de (Alt: Reichelt.de)	Taster 12x12 + USB-C + LEDs + Klemmen	**1 Set**	€4.50 (Interface Pack)	Board-level interface: SW_CAL_IN/OUT, LEDs, SW_PWR, USB-C entry, screw terminals & sockets.
Σ	**GRAND TOTAL COMPLETE PROTOTYPE HARDWARE (GROSS INCL. 19% VAT)**	**Consolidated Sourcing**	*Master Full BoM*	**16×**	**€70.77** **(Pure Parts Total)**	*Complete hardware BoM across all 13 line items before shipping.*

**** NOTE: BoM not final — Preliminary prototype specification. Component models, vendors, and prices subject to change during laboratory testing.***